

Introduction

Objective

Course objective – learn to design and develop Windows-based business applications using Visual Basic.NET programs that meet commercial programming standards.

- Program design and coding is satisfactory.
- Work is equivalent to that expected from someone already working in the information technology field as a professional programmer.
- Grade you according to commercial standards.

VB supports programming projects that run in both:

- Microsoft Windows and,
- Web (Internet) environments.

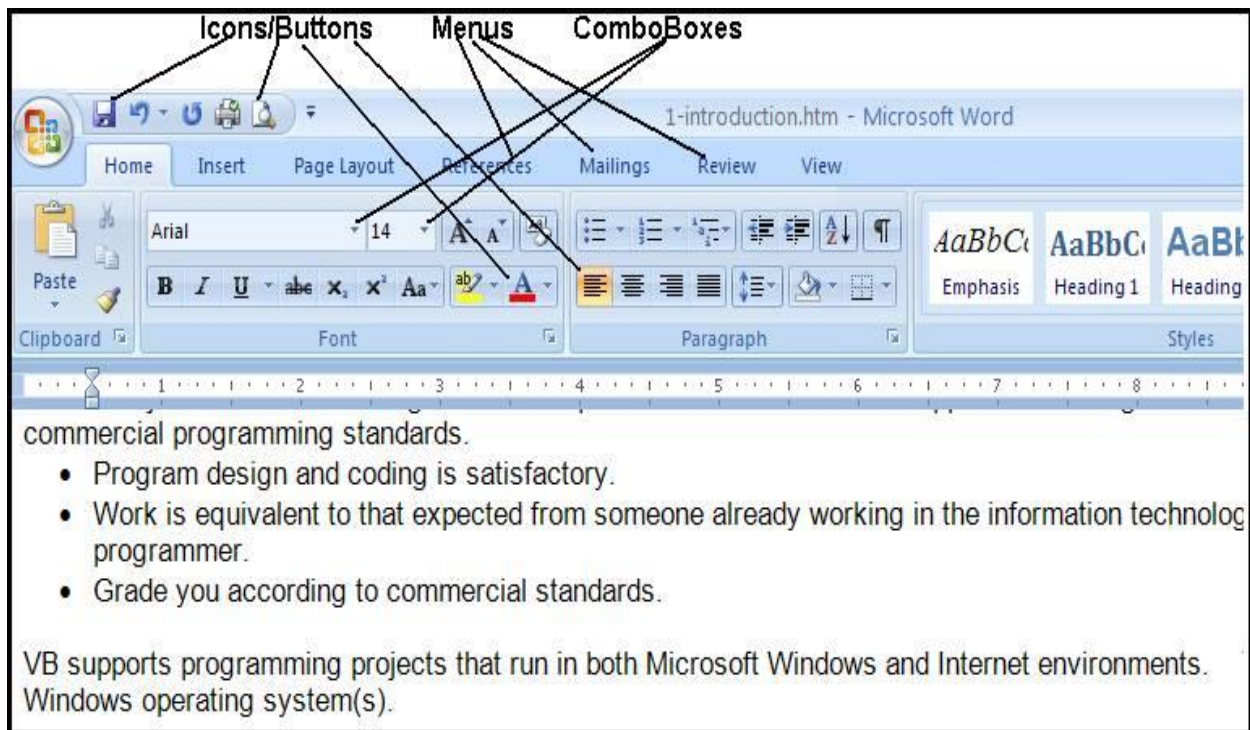
This course focuses on the Windows operating system(s).

Commercial GUI Design/ Development Approach

Modern programs run in a **graphical user interface (GUI)** environment. You can open up any desktop program or web browser and examine the **GUI**. This interface includes:

- Buttons (some with Icons and Graphical Images) and Menus on which you point and click.
- TextBox controls in which you type information.
- Graphical images to guide a system user in using the program.

A GUI for a version of Microsoft Word is shown in this figure:



Microsoft Visual Studio and Visual Basic (.NET Version)

Programming Languages

Microsoft's **Visual Studio** (also called **Visual Studio.NET**) includes several different programming languages:

- **Visual Basic**,
- **Visual C#** (C sharp),
- **Visual C++** (cee-plus-plus),
- **Visual F#** (F sharp),
- **JScript** and
- Web Development (called **ASP.NET**).
- It also includes the **.NET 4 Framework** upon which these languages operate.

All of these languages **compile**.

- This means they are translated from **human readable-form** to **machine readable-form** to the same **Microsoft Intermediate Language (MSIL)**.

Visual Studio Components

When Visual Studio is installed on a computer, there are two mandatory components to the installation and an optional third component.

- **NET Framework Class Library.** This is a library of predefined class objects. It enables you to quickly build a computer application through the use of predefined objects such as forms, text boxes, labels, buttons, drop-down list controls, and others (mandatory).
- **Common Language Runtime (CLR).** This component manages the execution of a programming project written in any of the languages that are included within Visual Studio including Visual Basic as a language (mandatory). This component is installed as part of the .NET Framework.
- **MSDN (Help).** This is the help component and provides access to a help reference library. This is covered in detail at the end of this set of notes. It is an optional, but highly recommended component.

Object-Oriented Programming Terminology

VB is an ***object-oriented programming language***.

- Means you work with **objects** in building an application.

Examples: Form objects, Button objects, TextBox objects, Label objects, ListBox objects, PictureBox objects, and more.

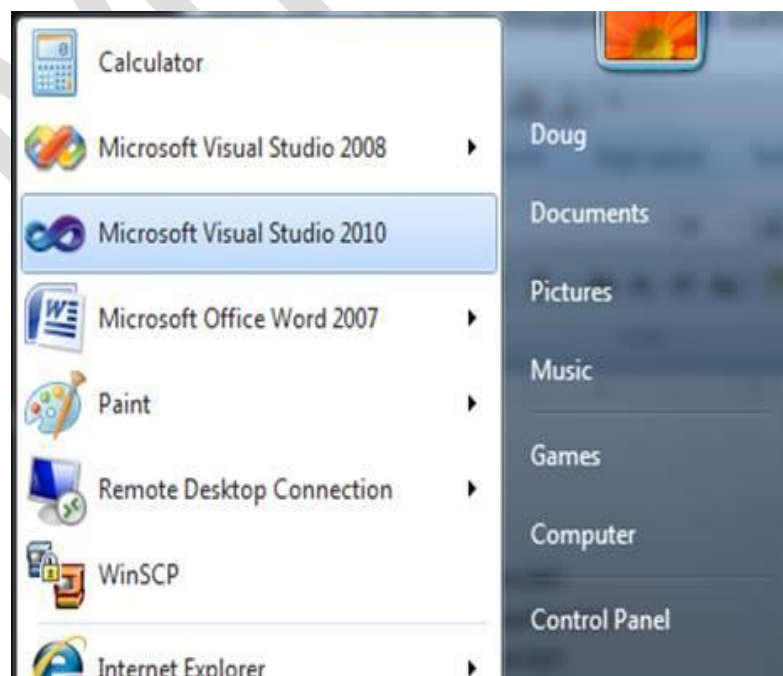
VB is also termed an ***event-driven programming language*** because you will write program code that responds to events that are controlled by the system user. Example events include:

- Clicking a button or menu.
- Opening or Closing a form.
- Moving the mouse over the top of an object such as a text box.
- Moving from one text box to another.

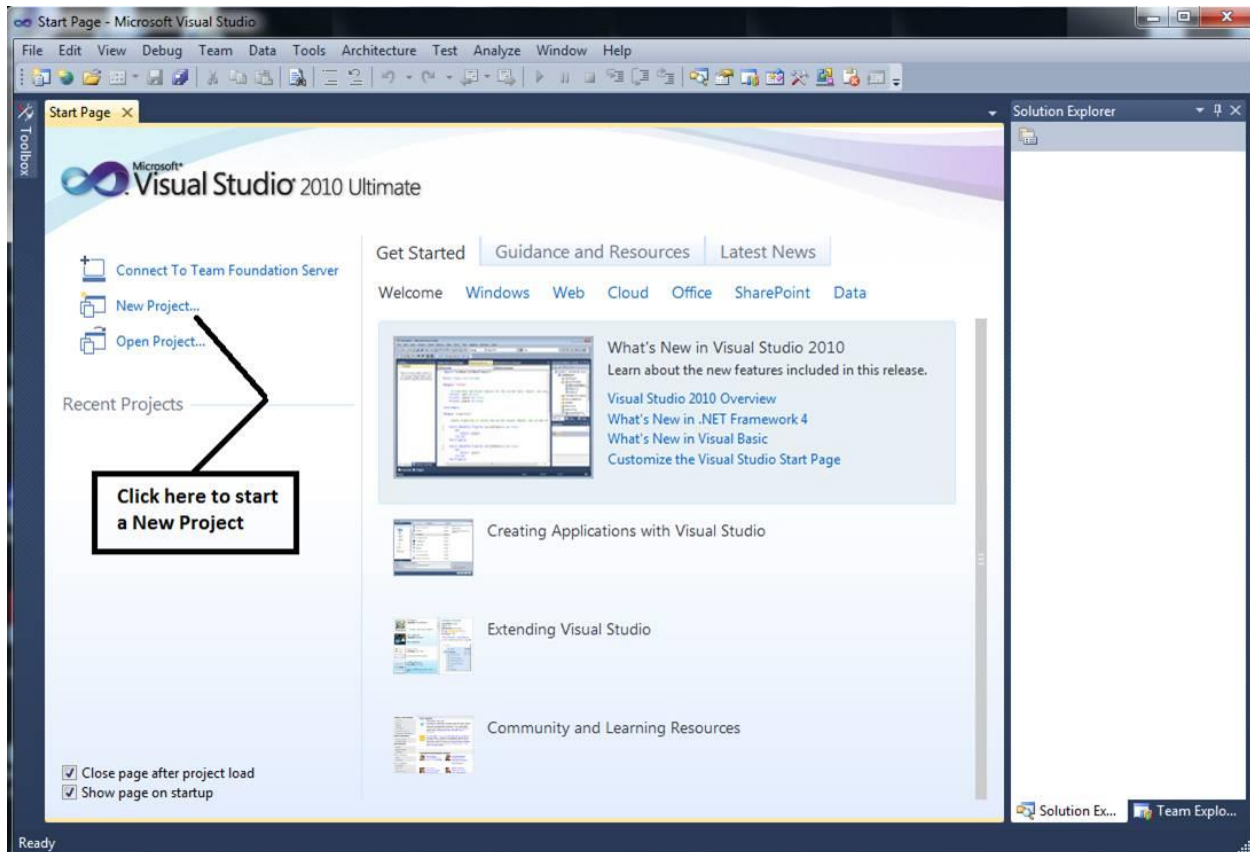
In order to work with VB, you need to understand "**object**" terminology as defined in Table 1.

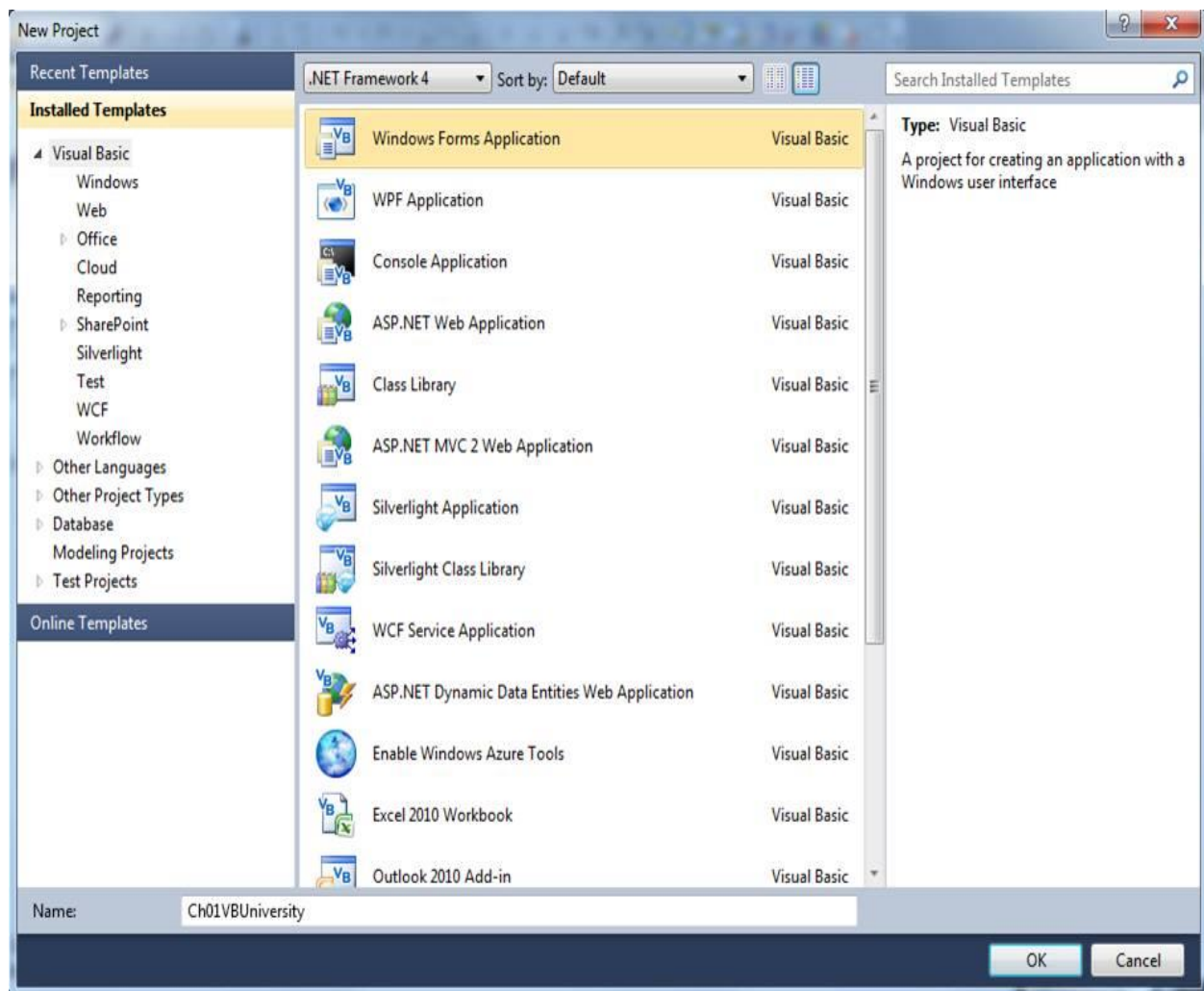
Table 1	
Terminology	Definition
Object	A thing – like a noun in English. Examples include forms and controls you place on forms such as buttons , text boxes , and icons .
Property	Objects have properties – like adjectives in English. Properties describe object behaviors. Examples of properties include Text , Name , BackColor , Font , and Size . Refer to a property by the notation ObjectName.PropertyName (use the .dot notation) – example: TotalDueTextBox.Text or AccountLabel.ForeColor .
Method	Like a verb in English – these are the actions that objects exhibit. Examples include methods to Show and Hide forms and methods to Print and Close forms. Refer to a method with the notation ObjectName.MethodName – example Me.Close will close the current form.
Event	Events are actions usually triggered by the system user such as clicking a button; however, events can also be triggered by the actions of objects. For example, closing a form can trigger an event.
Class	This is a really abstract term – it is a sort of template for an object. For example, all forms belong to the Form class of object. All buttons belong to the Button class of object. Classes include definitions for object properties, methods, and associated events. Each class is assigned an identifying namespace within the .NET Framework Class Library. Each new object you create is defined based on its class – the new object is called a class instance .

Start VB – Click the Windows **Start** button and locate **Microsoft Visual Studio 2010** – launch the program by clicking. The option will look something like this.



This is the Start Page for **Visual Basic 2010 Express Edition**.



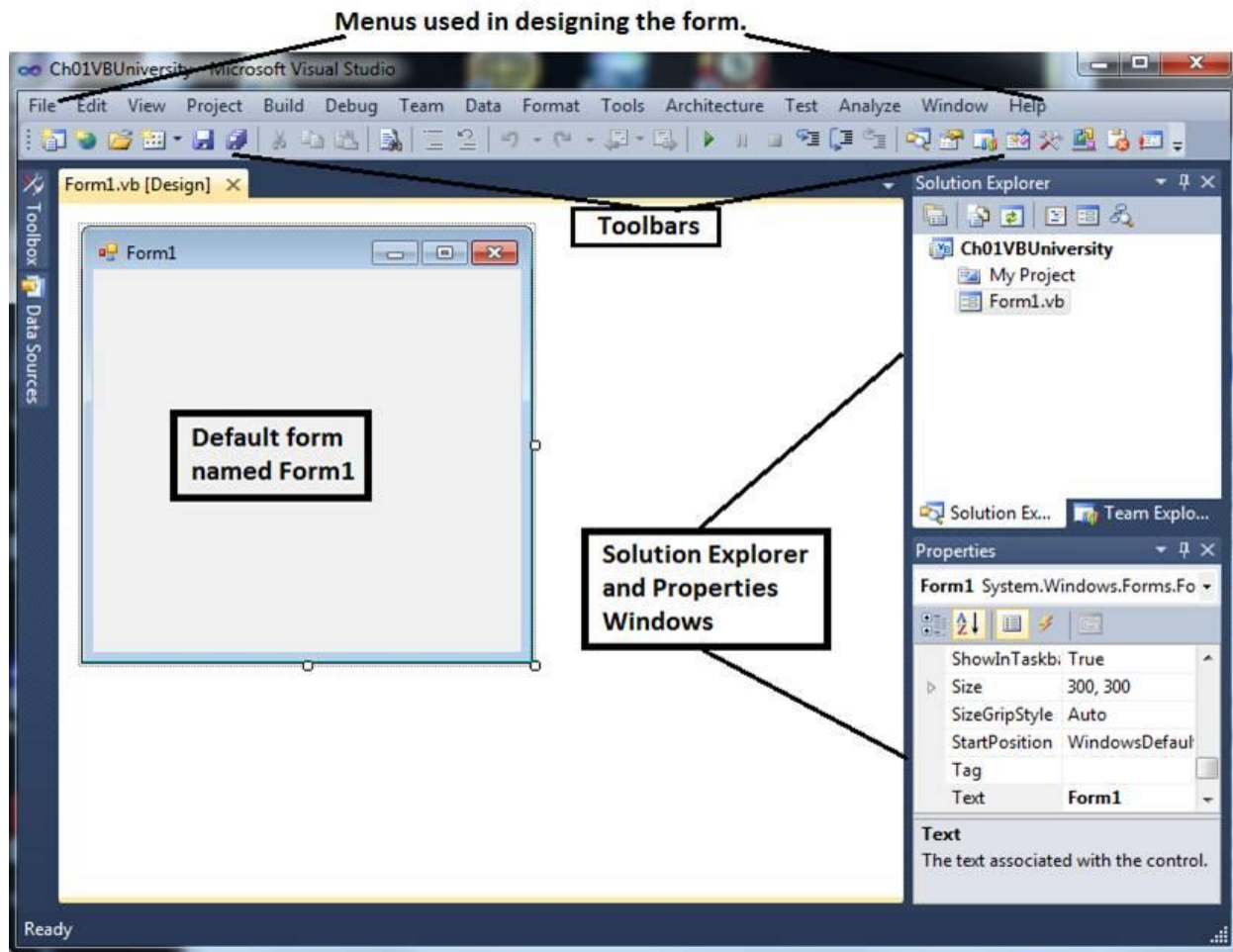


- Your first project will be a **Visual Basic Project** using a **Windows Forms Application** template as shown in the figures above.
- The default name is **WindowsApplication1** not a very useful name. Change the project name as you like.
- Click the **OK** button – it takes several seconds to create the project files from the template.

The Integrated Development Environment

Familiarize yourself with the **Integrated Development Environment (IDE)**.

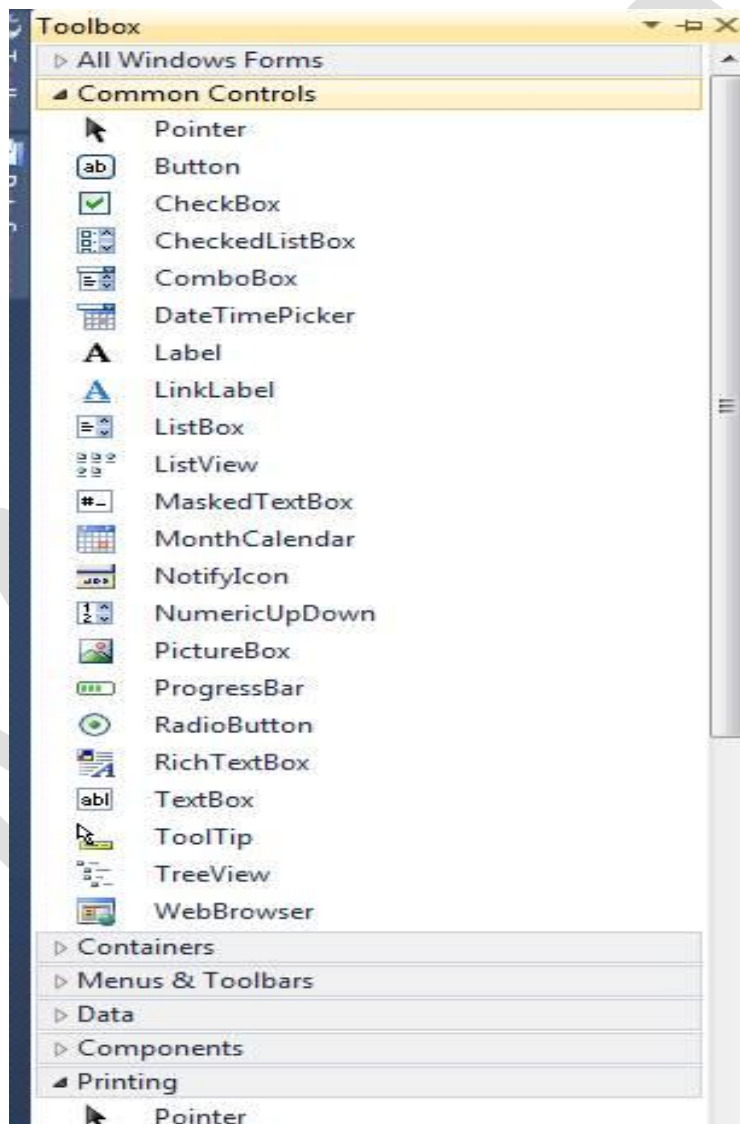
- Across the top are menus with different options used in the designing of an application.
- **Toolbars** with shortcut icons are shown below the menus.
- **Form Designer** (also termed the **Document Window**).
 - Displays open documents such as the **Form1** shown in the figure below.
 - Tabs along the top are used to allow you to switch rapidly between open documents.
 - Every project starts with a default form named **Form1** – the form you modify to build your business application.
 - The form can be resized by using the sizing handles.
 - The form can be renamed (you'll do this later in an exercise).
 - **Solution Explorer Window** – displays filenames for files that comprise a project.
 - **Properties Window** – displays properties of the currently selected object – in the figure the properties displayed are those of the Form object.
 - **ToolBox Window** – this is shown along the left edge of the figure in collapsed display.



The Toolbox:

- Contains controls that are used to build forms.
- Used to add controls (tools) to a form by either double-clicking or dragging/dropping (your option on which to use).

You can also close/open these windows with the **View** menu.



Resetting the Default IDE Layout

Sometimes you startup VB and the IDE layout is not what you expect.

- Windows you want such as Solution Explorer, Toolbox, or Properties may be closed.
- The quick way to reset the layout is through the **Window** menu, **Reset Window Layout** submenu option as shown in this figure.

